Section 1

```
(.venv) PS C:\Users\galnu\dcc-assignment2> python -m pytest tests/test_cap.py -v

============================================= test session starts =============================================

platform win32 -- Python 3.14.7, pytest-9.1.1, pluggy-1.6.0 -- C:\Users\galnu\dcc-assignment2\.venv\Scripts\python.exe

cachedir: .pytest_cache

rootdir: C:\Users\galnu\dcc-assignment2

configfile: pytest.ini

collected 21 items


tests/test_cap.py::test_majority_of_three_is_two PASSED                                                [  4%]

tests/test_cap.py::test_quorum_intersection_matches_r_plus_w_gt_n[3-1-1] PASSED                        [  9%]

tests/test_cap.py::test_quorum_intersection_matches_r_plus_w_gt_n[3-1-2] PASSED                        [ 14%]

tests/test_cap.py::test_quorum_intersection_matches_r_plus_w_gt_n[3-2-1] PASSED                        [ 19%]

tests/test_cap.py::test_quorum_intersection_matches_r_plus_w_gt_n[3-2-2] PASSED                        [ 23%]

tests/test_cap.py::test_quorum_intersection_matches_r_plus_w_gt_n[3-1-3] PASSED                        [ 28%]

tests/test_cap.py::test_quorum_intersection_matches_r_plus_w_gt_n[3-3-1] PASSED                        [ 33%]

tests/test_cap.py::test_quorum_intersection_matches_r_plus_w_gt_n[5-3-3] PASSED                        [ 38%]

tests/test_cap.py::test_quorum_intersection_matches_r_plus_w_gt_n[5-2-3] PASSED                        [ 42%]

tests/test_cap.py::test_quorum_intersection_matches_r_plus_w_gt_n[5-2-4] PASSED                        [ 47%]

tests/test_cap.py::test_cp_write_succeeds_on_all_healthy_nodes PASSED                                  [ 52%]

tests/test_cap.py::test_cp_write_to_majority_partition_succeeds PASSED                                 [ 57%]

tests/test_cap.py::test_cp_write_to_minority_partition_is_rejected PASSED                              [ 61%]

tests/test_cap.py::test_ap_write_to_minority_partition_is_accepted PASSED                              [ 66%]

tests/test_cap.py::test_ap_divergent_writes_converge_by_lww_after_heal PASSED                          [ 71%]

tests/test_cap.py::test_lww_decides_by_timestamp_not_by_arrival_order PASSED                           [ 76%]

tests/test_cap.py::test_linearizable_read_through_follower_sees_latest_cp_write PASSED                 [ 80%]

tests/test_cap.py::test_r_plus_w_not_greater_than_n_allows_stale_read PASSED                           [ 85%]

tests/test_cap.py::test_linearizable_read_unavailable_but_follower_read_available_in_partition PASSED  [
90%]

tests/test_cap.py::test_follower_read_beyond_max_staleness_is_rejected PASSED                          [ 95%]

tests/test_cap.py::test_duplicate_idempotency_key_is_not_applied_twice PASSED                          [100%]


============================================== 21 passed in 1.31s ==============================================
```

(.venv) PS C:\Users\galnu\dcc-assignment2> python client.py kvput q1 v1 --strategy cp

OK success=True acks_received=2

(.venv) PS C:\Users\galnu\dcc-assignment2> python client.py kvput q2 v2 --strategy cp --addr 127.0.0.1:50053

FAILED acks_received=1 error=ReplicaUnavailableError: quorum lost (1/3 replicas reachable, need 2)

(.venv) PS C:\Users\galnu\dcc-assignment2> python client.py kvput q3 red --strategy ap --ts 200 --addr 127.0.0.1:50053

OK success=True acks_received=1

(.venv) PS C:\Users\galnu\dcc-assignment2> python client.py kvput q3 blue --strategy ap --ts 100

OK success=True acks_received=2

(.venv) PS C:\Users\galnu\dcc-assignment2> python client.py kvpartition --addr 127.0.0.1:50051

partition on 127.0.0.1:50051: blocked=[]

(.venv) PS C:\Users\galnu\dcc-assignment2> python client.py kvpartition --addr 127.0.0.1:50052

partition on 127.0.0.1:50052: blocked=[]

(.venv) PS C:\Users\galnu\dcc-assignment2> python client.py kvpartition --addr 127.0.0.1:50053

partition on 127.0.0.1:50053: blocked=[]

(.venv) PS C:\Users\galnu\dcc-assignment2> python client.py kvget q3 --consistency stale --addr 127.0.0.1:50051

value=red read_timestamp=200 is_stale=False

(.venv) PS C:\Users\galnu\dcc-assignment2> python client.py kvget q3 --consistency stale --addr 127.0.0.1:50052

value=red read_timestamp=200 is_stale=False

(.venv) PS C:\Users\galnu\dcc-assignment2> python client.py kvget q3 --consistency stale --addr 127.0.0.1:50053

value=red read_timestamp=200 is_stale=False

(.venv) PS C:\Users\galnu\dcc-assignment2>

Scenario	Reachable replicas	Expected	Actual
CP write to DC-1 (A, B)	2 of 3	success	success=True, acks_received=2
CP write to DC-2 (C)	1 of 3	rejected	FAILED, ReplicaUnavailableError (1/3 reachable, need 2)
AP write to DC-2 (C)	1 of 3	success	success=True, acks_received=1
AP after partition heals	3 of 3	higher timestamp wins	red (ts=200) on all three replicas

Section 2

(.venv) PS C:\Users\galnu\dcc-assignment2> python perf_cap_bench.py

Cluster: 3 replicas on localhost, client targets Replica C (follower), 1000 reads per scenario, max_staleness_ms = 5000

| Read Mode | Target Node | Median Latency (ms) | p95 Latency (ms) | Read Availability (%) |

|---|---|---|---|---|

| Linearizable | Leaseholder (Remote) | 0.69 | 0.90 | 100.0 |

| Follower Stale | Local Follower | 0.33 | 0.43 | 100.0 |

During a partition (C isolated from A and B), 200 reads per scenario:

| Read Mode | Target Node | Read Availability (%) |

|---|---|---|

| Linearizable | Leaseholder (unreachable) | 0.0 |

| Follower Stale | Local Follower | 100.0 |

Section 3

1. If R + W > N, the replicas that wrote and the replicas that read must overlap: at least one replica is in both groups. This replica has the latest write, so the read is always fresh (linearizable). If R + W ≤ N, the groups may not overlap, so a read can go to replicas without the new write and return an old value. In my test: W = 2 (A, B), R = 1 (C), 2 + 1 = 3, not more than 3, so C returned an old value.
2. With QUORUM_CP, the write on the minority node C was rejected because it could not reach a majority (consistency over availability). With ALWAYS_AP, the write was accepted (availability over consistency), but the replicas had different values until the partition healed. Then Last-Write-Wins kept the newer value (red, ts = 200) and the older one (blue, ts = 100) was lost.